

# Stochastic Hydrology – Semester Project Description

The semester project for the Stochastic Hydrology course will be graded on two components:

- Report 30 % of final grade
- Recorded Presentation 10 % of final grade

The individual performance assessments do not all have to be passed individually. For equal contribution to the semester project, all group members get the same grade. To assess this, students will be asked to report the share each member contributed at the end of the semester. All materials have to be submitted until 9. June 2026, 21:00.

## Task

The project is carried out in groups of two (2) students. Each group is required to identify and obtain its own dataset and to formulate independent research questions based on the selected data. The analysis must make use of methods introduced in at least two different course modules. In addition, the project should include at least one method or aspect that has not been explicitly covered during the course, thereby encouraging independent exploration beyond the lecture material.

## Report

Requirements:

- Typed, submitted as a pdf document
- Maximum length of 10 pages, A4 (excl. title page and references)
- Font size 11 pt., single spaced

The report should include the following sections:

- *Abstract:*  
Max. 200 words. This section should provide a pertinent overview of the work.  
Example flow for your abstract: place the question addressed in a broad context and highlight the purpose of the study; briefly describe the main methods applied; summarize your project's main findings; finally, indicate the main conclusions or interpretations.
- *Introduction:*  
Briefly place the study in a broad context and highlight why it is important. Here, you should also define the purpose of your project and its significance. Briefly review the cur-

rent state of the research field and cite key publications. As far as possible, please keep the introduction comprehensible to scientists outside your particular field of research.

- *Research questions:*  
One or several research questions you want to address as part of your project.
- *Hypotheses:*  
Formulate a research hypothesis for each of the research questions to be tested as part of your analysis.
- *Data:*  
Provide a concise and complete description of the data used for the analysis.
- *Methods:*  
Give a brief overview of the methods you will be using in your analysis. For this report we don't need the methods re-explained, just provide some information on the main steps.
- *Results:*  
Provide a concise and precise description of your results.
- *Discussion:*  
Discuss the results and how they can be interpreted in perspective of previous studies and of your initial hypotheses. Limitations of this study and future research directions may also be highlighted.
- *Conclusions:*  
Summarize the main conclusions of your work.
- *References:*  
List the articles and resources cited in your report.

## **Recorded Presentation**

The recorded presentation has to fulfil the following criteria:

- 10 minutes
- Recording of sound and slides is required (e.g., through zoom recording).
- Each group member has to present a part of the presentation.
- The presentation should cover the main aspects of the report, including a introduction, research questions and hypothesis, a description of the data and methods, the results, discussion and conclusion.